

hello guys !
I am Muskan Garg



HOSPITAL MANAGEMENT DATA ANALYSIS THROUGH SQL





Q1) CREATE A DASHBOARD SHOWING THE TOTAL NUMBER OF PATIENTS, DOCTORS , APPOINTMENTS, TREATMENTS AND TOTAL REVENUE GENERATED BY THE HOSPITAL.



```
SELECT (SELECT COUNT(patient_id) FROM patients_data) AS patient_count,  
(SELECT COUNT(appointment_id) FROM appointments_record) As appointment_count,  
(SELECT COUNT(doctor_id) FROM doctors_record) AS doctor_count,  
(SELECT COUNT(treatment_id) FROM treatment_record) AS treatment_count,  
(SELECT SUM(amount) FROM billing_record) AS total_revenue;
```

	patient_count bigint	appointment_count bigint	doctor_count bigint	treatment_count bigint	total_revenue numeric
1	50	200	10	200	551249.85



Q2) ANALYZE THE MONTHLY REVENUE TREND AND IDENTIFY THE MONTH WITH THE HIGHEST AND LOWEST REVENUE.



```
SELECT MIN(monthly_revenue) AS lowest_revenue,MAX(monthly_revenue) AS highest_revenue
FROM(SELECT EXTRACT(Month FROM bill_date) AS bill_month ,
SUM(amount) AS monthly_revenue
FROM billing_record
GROUP BY bill_month) AS revenue_details;
```



	lowest_revenue numeric	highest_revenue numeric
1	27569.71	64271.54

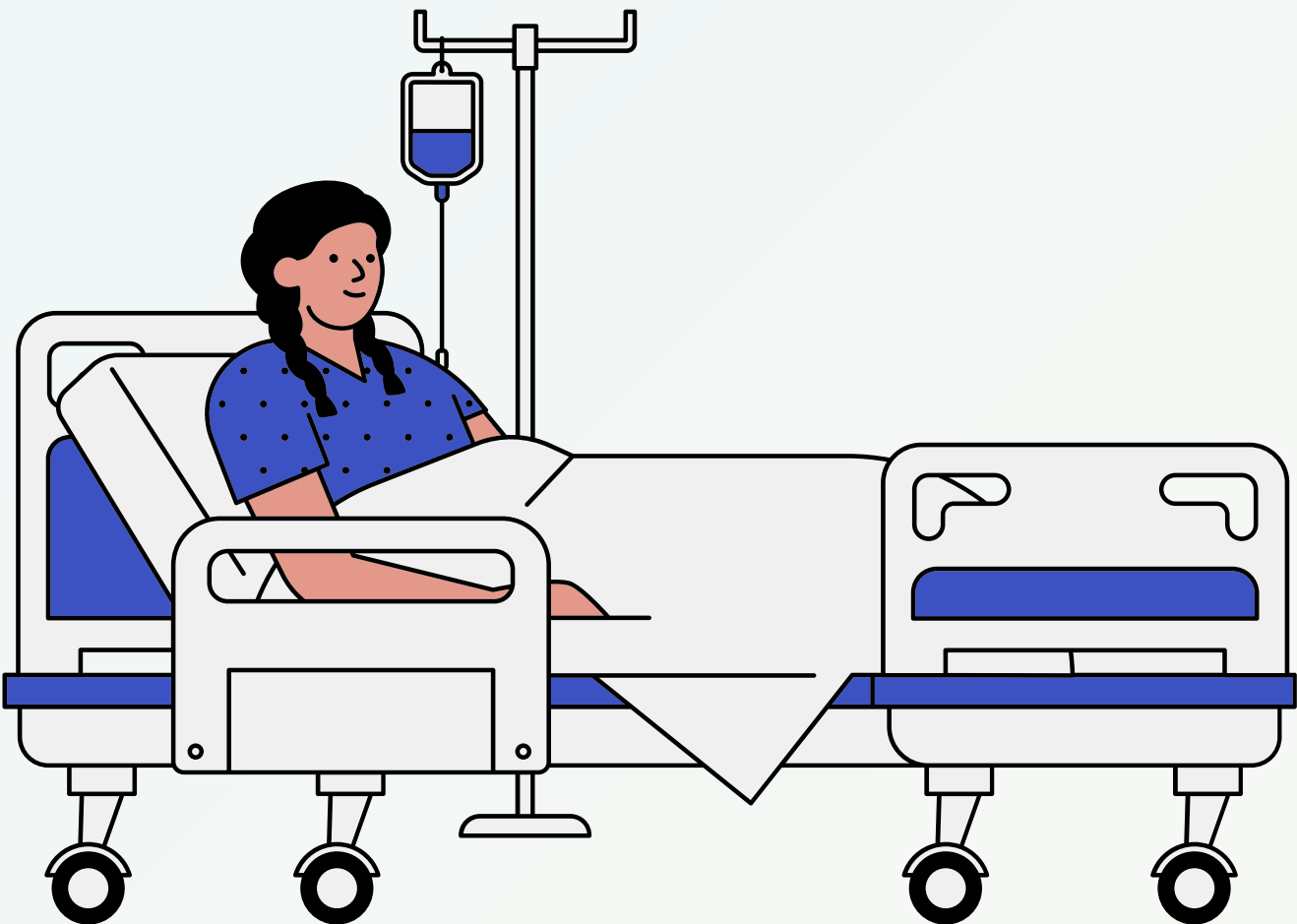




Q3) IDENTIFY THE TOP 10 PATIENTS BASED ON THEIR TOTAL SPENDING ON TREATMENTS AND BILLING.



```
SELECT p.patient_id,p.first_name,p.last_name,(b.amount+t.cost) AS total_amount_spend
FROM patients_data p
INNER JOIN
billing_record b
on p.patient_id=b.patient_id
inner join
treatment_record t
on b.treatment_id=t.treatment_id
order by total_amount_spend desc limit 10;
```



	patient_id [PK] character varying (500)	first_name character varying (500)	last_name character varying (500)	total_amount_spend numeric
1	P024	Sarah	Brown	9947.26
2	P017	Jane	Jones	9932.36
3	P021	Michael	Wilson	9929.42
4	P050	Laura	Wilson	9921.30
5	P044	Robert	Taylor	9890.06
6	P047	Jane	Moore	9780.50
7	P038	David	Smith	9692.40
8	P029	David	Smith	9668.04
9	P033	Michael	Wilson	9666.34
10	P049	David	Moore	9618.62

Q4) FIND THE TOP 5 DOCTORS WHO GENERATED THE HIGHEST REVENUE FOR THE HOSPITAL.

```
SELECT d.doctor_id,d.first_name,d.last_name,SUM(b.amount) AS revenue
FROM doctors_record d
INNER JOIN
appointments_record a
ON d.doctor_id=a.doctor_id
inner join
billing_record b
ON a.patient_id=b.patient_id
GROUP BY d.doctor_id
ORDER BY revenue DESC LIMIT 5;
```



	doctor_id [PK] character varying (500) 	first_name character varying (500) 	last_name character varying (500) 	revenue numeric 
1	D005	Sarah	Taylor	394356.89
2	D006	Alex	Davis	339280.84
3	D001	David	Taylor	336851.92
4	D010	Linda	Wilson	304846.08
5	D003	Jane	Smith	279957.14



Q5) COMPARE THE NUMBER OF APPOINTMENTS HANDLED BY EACH DOCTOR AND RANK THEM FROM HIGHEST TO LOWEST.



```
SELECT d.doctor_id,d.first_name,d.last_name,
COUNT(a.appointment_id) AS count_appointments,
RANK() OVER(ORDER BY COUNT(a.appointment_id)) AS doctor_rank
FROM doctors_record d
INNER JOIN
appointments_record a
ON d.doctor_id=a.doctor_id
GROUP BY d.doctor_id
ORDER BY doctor_rank DESC;
```



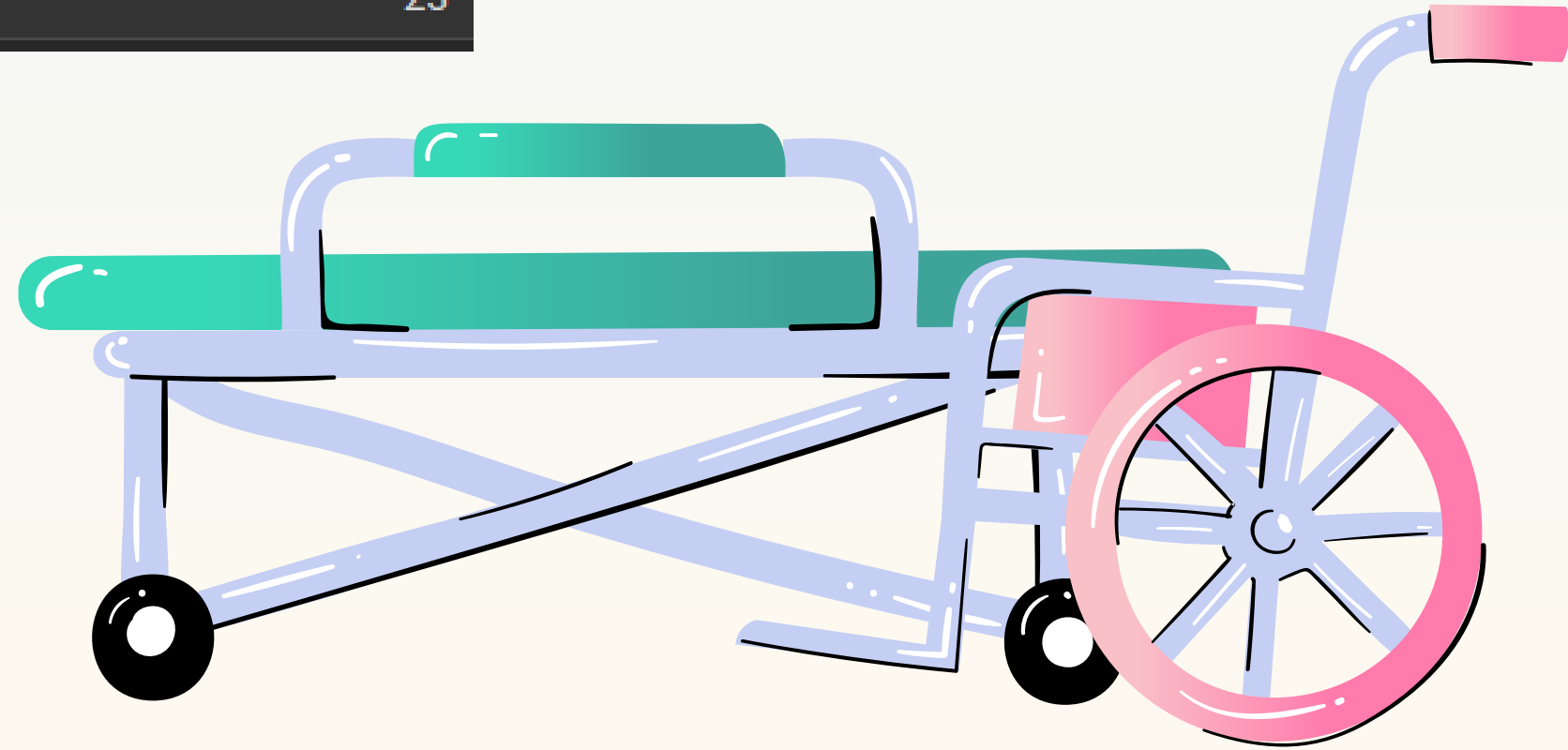
	doctor_id [PK] character varying (500)	first_name character varying (500)	last_name character varying (500)	count_appointments	doctor_rank
1	D005	Sarah	Taylor	29	10
2	D001	David	Taylor	25	9
3	D006	Alex	Davis	24	8
4	D003	Jane	Smith	22	7
5	D002	Jane	Davis	21	6
6	D010	Linda	Wilson	19	5
7	D009	Sarah	Smith	17	4
8	D008	Linda	Brown	16	3
9	D004	David	Jones	14	2
10	D007	Robert	Davis	13	1

Q6) DETERMINE WHICH DOCTOR HAS TREATED THE HIGHEST NO. OF UNIQUE PATIENTS.



```
SELECT d.doctor_id,d.first_name,d.last_name,  
COUNT(DISTINCT a.patient_id) AS count_unique_patient  
FROM doctors_record d  
INNER JOIN  
appointments_record a  
ON d.doctor_id=a.doctor_id  
GROUP BY d.doctor_id  
ORDER BY count_unique_patient DESC LIMIT 1;
```

	doctor_id [PK] character varying (500)	first_name character varying (500)	last_name character varying (500)	count_unique_patient bigint
1	D005	Sarah	Taylor	23





Q7) IDENTIFY THE MOST FREQUENTLY PERFORMED TREATMENT AND CALCULATE THE REVENUE GENERATED FROM EACH TREATMENT TYPE.

```
SELECT treatment_type,COUNT(treatment_type) AS treatment_frequency,  
SUM(cost) AS treatment_cost FROM treatment_record  
GROUP BY treatment_type  
ORDER BY treatment_frequency DESC;
```



	treatment_type character varying (500)	treatment_frequency bigint	treatment_cost numeric
1	Chemotherapy	49	128855.68
2	X-Ray	41	110653.67
3	ECG	38	96224.24
4	MRI	36	116098.16
5	Physiotherapy	36	99418.10



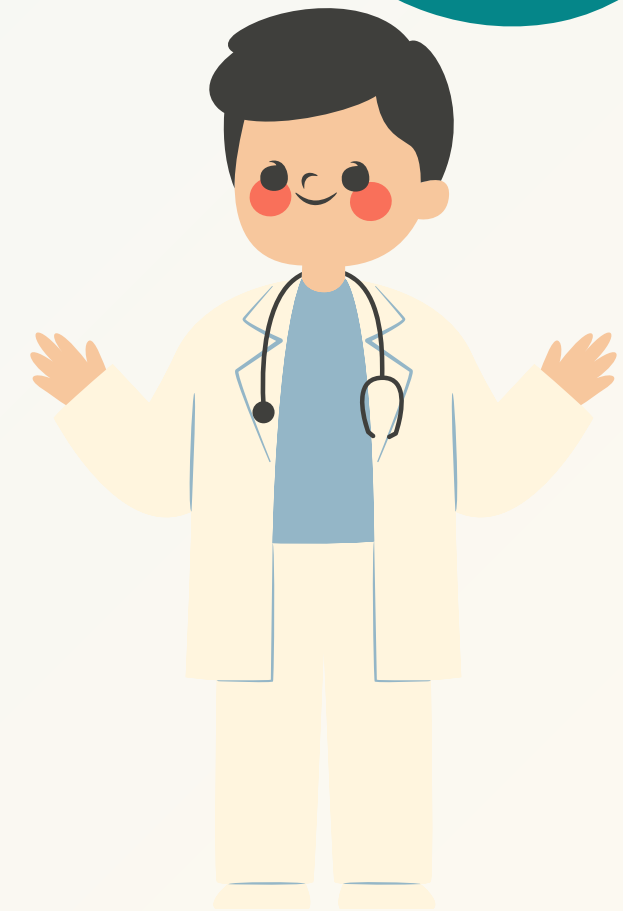


Q8) ANALYZE APPOINTMENT STATUS BY CALCULATING THE PERCENTAGE OF COMPLETED, CANCELLED AND PENDING APPOINTMENTS.



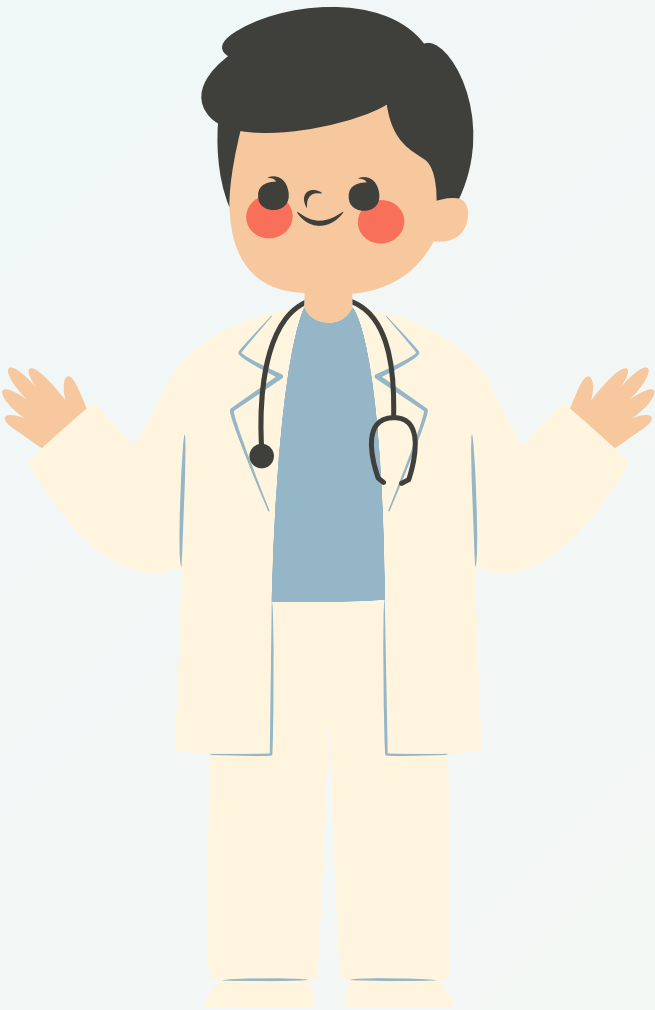
```
SELECT status,COUNT(status)*100/SUM(COUNT(status)) OVER()  
AS appointment_status_percentage  
FROM appointments_record  
GROUP BY status;
```

	status character varying (500) 	appointment_status_percentage numeric 
1	Scheduled	25.500000000000000000
2	No-show	26.000000000000000000
3	Completed	23.000000000000000000
4	Cancelled	25.500000000000000000



Q9) FIND THE BUSIEST DAY AND BUSIEST MONTH BASED ON THE NUMBER OF APPOINTMENTS SCHEDULED.

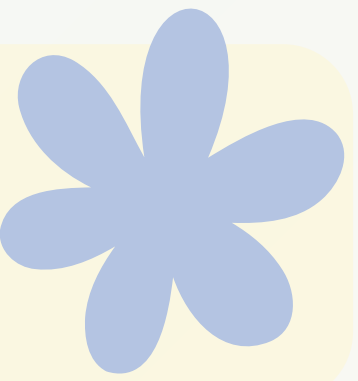
```
SELECT appointment_date,EXTRACT(MONTH FROM appointment_date) AS month,
EXTRACT(day FROM appointment_date) AS day,
COUNT(appointment_id) AS total_appointments
FROM appointments_record
GROUP BY appointment_date
ORDER BY total_appointments DESC LIMIT 1;
```



	appointment_date 	month 	day 	total_appointments 
	date	numeric	numeric	bigint
1	2023-08-16	8	16	5



Q10) GENERATE A SPECIALIZATION WISE REPORT SHOWING THE TOTAL NUMBER OF DOCTORS, APPOINTMENTS, PATIENTS TREATED AND REVENUE GENERATED.



```
SELECT d.specialization,COUNT(t.treatment_id) AS treatment_count,  
SUM(b.amount) AS revenue_generated,COUNT(DISTINCT b.patient_id) AS treated_patient_count,  
COUNT(a.appointment_id) AS appointment_count,  
COUNT(d.doctor_id) AS doctor_count FROM doctors_record d  
inner join appointments_record a  
on d.doctor_id=a.doctor_id  
INNER JOIN billing_record b  
ON b.patient_id=a.patient_id  
INNER JOIN treatment_record t  
ON t.treatment_id=b.treatment_id  
group by d.specialization;
```

	specialization character varying (500) 🔒	treatment_count bigint 🔒	revenue_generated numeric 🔒	treated_patient_count bigint 🔒	appointment_count bigint 🔒	doctor_count bigint 🔒
1	Dermatology	353	965566.84	37	353	353
2	Oncology	177	493428.71	22	177	177
3	Pediatrics	472	1289294.61	47	472	472



Q11) ANALYZE PAYMENT METHODS BY FINDING THE TOTAL Revenue AND PERCENTAGE CONTRIBUTION OF EACH PAYMENT METHOD (CASH, UPI, CARD ETC.)



```
SELECT payment_method,SUM(amount)*100/SUM(SUM(amount)) OVER() AS percentage_contribution,  
SUM(amount) AS revenue_generated FROM billing_record  
GROUP BY payment_method;
```



	payment_method character varying (500)	percentage_contribution numeric	revenue_generated numeric
1	Insurance	33.0449577446597038	182160.28
2	Credit Card	36.5319700313750652	201382.43
3	Cash	30.4230722239652310	167707.14



Q12) RANK DOCTORS WITH EACH SPECIALIZATION BASED ON THE REVENUE THEY GENERATED USING WINDOW FUNCTIONS.



```
SELECT d.specialization,sum(b.amount) AS revenue_generated,  
RANK() OVER(order by sum(b.amount) DESC ) AS patient_rank  
FROM doctors_record d inner join appointments_record a  
on d.doctor_id=a.doctor_id  
inner join billing_record b  
on a.patient_id=b.patient_id  
group by d.specialization;
```

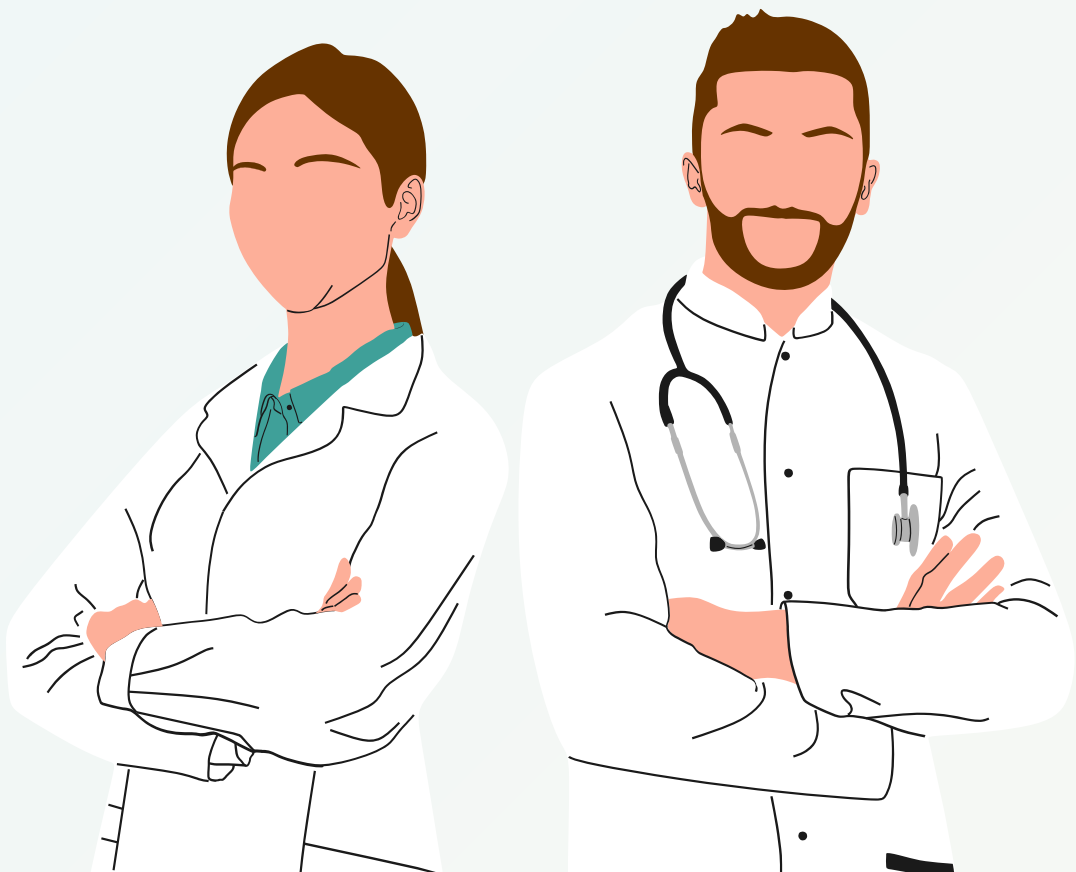
	specialization character varying (500) 	revenue_generated numeric 	patient_rank bigint 
1	Pediatrics	1289294.61	1
2	Dermatology	965566.84	2
3	Oncology	493428.71	3



Q13) GENERATE A MONTHLY FINANCIAL REPORT SHOWING TOTAL NUMBER BILLS, TOTAL REVENUE AND AVERAGE BILL AMOUNT.



```
SELECT EXTRACT(MONTH FROM bill_date) AS payment_month,
SUM(amount) AS total_revenue,AVG(amount) AS average_revenue,COUNT(bill_id) AS total_bills
FROM billing_record
GROUP BY payment_month;
```



	payment_month numeric	total_revenue numeric	average_revenue numeric	total_bills bigint
1	3	47304.29	2489.6994736842105263	19
2	1	58701.23	2935.0615000000000000	20
3	4	64271.54	2570.8616000000000000	25
4	9	33426.53	3038.7754545454545455	11
5	6	56887.82	3160.4344444444444444	18
6	10	43314.15	3093.8678571428571429	14
7	8	41958.67	2797.2446666666666667	15
8	7	39880.19	2492.5118750000000000	16
9	2	36669.69	2619.2635714285714286	14
10	11	52474.98	3086.7635294117647059	17
11	12	27569.71	2297.4758333333333333	12

THANK YOU

